

Assessment of clinical competencies in medical education in the era of Artificial Intelligence: an integrative literature review

Avaliação das competências clínicas na educação médica na era da Inteligência Artificial: uma revisão integrativa da literatura

Evaluación de las competencias clínicas en la educación médica en la era de la Inteligencia Artificial: una revisión integrativa de la literatura

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ABSTRACT: Artificial intelligence (AI) has rapidly expanded within medical education, challenging traditional approaches to assessing clinical competence. Competency-based curricula require assessment strategies capable of inferring professional performance, clinical reasoning, communication, and decision-making in authentic clinical contexts. However, written examinations and case-based assignments, historically used as proxies for competence, are increasingly affected by AI systems capable of generating clinically structured responses. This transformation raises concerns regarding the validity of assessment evidence and the interpretation of learner performance. This integrative literature review aimed to analyze how clinical competence assessment in medical education has been discussed in the context of AI. Searches were conducted in PubMed, Scopus, Web of Science, and ERIC, including studies published between 2019 and 2026. Twenty-four studies met the inclusion criteria. Data were synthesized through interpretive thematic analysis and categorized according to levels of evidence. The findings indicate that AI does not replace the human assessor but contributes to a hybrid assessment ecosystem. Performance-based methods, such as OSCEs, workplace-based assessments, and portfolios, gain relevance, while assessment shifts from verifying correct answers to evaluating reasoning processes and justification. AI-mediated feedback enhances individualized learning and longitudinal monitoring; however, risks include superficial performance, compromised authenticity, and challenges to professional autonomy. Ethical concerns involve authorship transparency, reliability, and patient safety. Overall, clinical competence assessment must evolve from knowledge recall toward evaluation of judgment in human-AI interaction. Future assessment models should explicitly incorporate critical use of AI as part of contemporary medical competence while preserving the centrality of clinical reasoning and structured faculty mediation.

Keywords: medical education; clinical competence; artificial intelligence; assessment; clinical reasoning.

1 INTRODUCTION

Medical education has increasingly been guided by competency-based curricula, in which assessment is intended to infer professional performance in clinical situations (Ten Cate *et al.*, 2021; Ten Cate & Taylor, 2021). Within this framework, assessment instruments must capture clinical reasoning, decision-making, and communication, moving beyond the mere verification of content knowledge (Jeffery *et al.*, 2024; Soledad *et al.*, 2025). Nevertheless,

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traditional written examinations remain widely used despite their limitations in representing clinical competence (Elsamanoundy *et al.*, 2024; Kassam *et al.*, 2024).

The recent diffusion of artificial intelligence systems capable of generating text and suggesting clinical management challenges the validity of these strategies (Kung *et al.*, 2023; Gilson *et al.*, 2023). Activities commonly employed to infer learning, such as case analyses and open-ended responses, may no longer necessarily reflect the student's cognitive processing, but rather technological mediation (Sallam, 2023). Consequently, artificial intelligence reshapes the very meaning of assessment evidence.

In response, medical education has expanded the use of performance-based assessments, including structured observation, simulation, and portfolios, particularly within the framework of programmatic assessment and Entrustable Professional Activities (Ten Cate *et al.*, 2021). However, these instruments emerge from distinct theoretical perspectives and lack an integrated synthesis regarding their capacity to support inferences of professional competence in environments permeated by automated cognitive technologies.

In this context, a critical integration of scientific literature on the assessment of clinical competencies in the presence of artificial intelligence becomes necessary. Therefore, this study aims to analyze, in the scientific literature, how the assessment of clinical competencies in medical education has been discussed in the context of artificial intelligence.

2 METHOD

This study consists of an Integrative Literature Review, a method that enables the systematic and comprehensive gathering and synthesis of research findings on a given topic, allowing a broader understanding of the investigated phenomenon. Grounded in the principles of Evidence-Based Practice, this approach aims to construct robust knowledge useful for both professional practice and educational policy development (Lemes *et al.*, 2021; Ganong, 1987; Lockwood, Munn, & Porritt, 2015).

The guiding research question was developed using the PICO strategy (Population, Interest, Context), considering a population composed of medical students and physicians in training, an interest related to the assessment of clinical competencies, and a context referring to the presence of artificial intelligence in educational processes (Stern, Jordan, & McArthur, 2014). Accordingly, the following question was defined: how has the scientific literature discussed the assessment of clinical competencies in medical education in the presence of artificial intelligence?

The search was conducted in the PubMed/MEDLINE, Scopus, Web of Science, and ERIC (Education Resources Information Center) databases using controlled descriptors and free terms in English combined with Boolean operators: ("Education, Medical" OR "Medical Education" OR "Internship and Residency" OR "medical students") AND ("Clinical Competence" OR "clinical skills" OR "clinical reasoning" OR "competency-based education") AND (OSCE OR "workplace-based assessment" OR Mini-CEX OR "performance assessment" OR "authentic assessment") AND ("Artificial Intelligence" OR "generative AI" OR chatbot OR "large language model").

Studies published between 2019 and 2026, available in full text in Portuguese, English, or Spanish, addressing the assessment of clinical competencies in medical education in connection with transformations arising from artificial intelligence were included. Studies conducted in primary education settings or not applicable to medical training contexts were excluded, as well as research focused exclusively on traditional cognitive performance, studies addressing technological development without pedagogical discussion, and editorials, letters to the editor, and conference abstracts.

The systematic search across the selected databases identified 186 studies in PubMed, 243 in Scopus, 121 in Web of Science, and 64 in ERIC, totaling 614 records. All records were exported to a reference manager for organization and duplicate removal.

During the screening stage, titles and abstracts of the 614 identified records were reviewed according to the eligibility criteria, resulting in 72 articles for full-text reading. After full analysis, 48 studies were excluded for not meeting the proposed methodological design, not directly addressing the thematic scope, or constituting non-relevant secondary studies. Ultimately, 24 articles composed the final review sample.

Study selection was conducted independently by two researchers responsible for the methodological stage, with disagreements resolved by consensus.

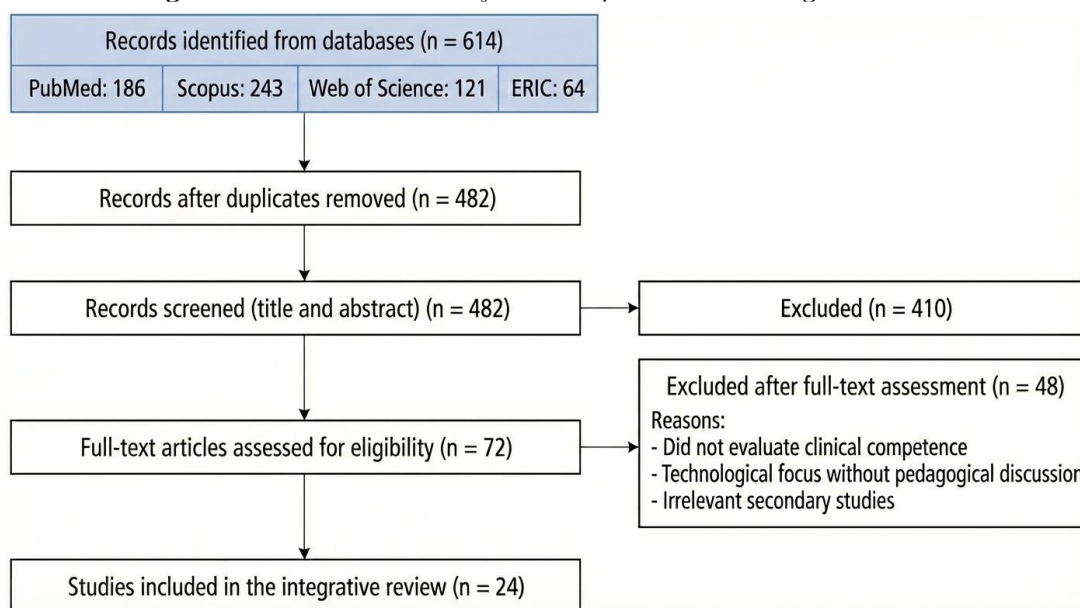
The included studies were classified according to levels of evidence based on Joanna Briggs Institute recommendations (Lockwood *et al.*, 2020): Level I – systematic reviews and randomized clinical trials; Level II – experimental studies; Level III – quasi-experimental studies; Level IV – observational studies; and Level V – qualitative or theoretical studies. This classification was used for methodological characterization and critical interpretation of findings and did not constitute an exclusion criterion.

After final selection, the articles were organized in a data extraction instrument containing authorship, year, country, training level, assessment instrument used, dimension of competence analyzed, and main results. Synthesis occurred through interpretive thematic

analysis, allowing the construction of analytical categories representing convergences and gaps in the investigated field. Results are presented descriptively and through a summary table.

Figure 1 presents the methodological pathway adopted in this Integrative Review, illustrating the stages of identification, selection, eligibility, and inclusion of studies.

Figure 1: Flowchart of the study selection process in this Integrative Review



Source: Prepared by the authors

3 RESULTS

After applying the search and selection criteria, 24 studies published between 2019 and 2026 were included, retrieved from the PubMed, Scopus, Web of Science, and ERIC databases. Most studies (75%) were conducted in English-speaking countries, particularly the United States, the United Kingdom, and Canada, followed by publications from Latin America, Europe, and Asia. Regarding methodological design, quantitative studies predominated (58%), followed by qualitative (25%) and mixed-methods studies (17%). The target population consisted mainly of undergraduate medical students (67%), as well as residents and faculty involved in assessment processes.

The analyzed studies addressed different interfaces between artificial intelligence and the assessment of clinical competence in medical education. The main investigated focuses included the use of automated feedback systems and clinical reasoning support (Kung *et al.*, 2023; Gilson *et al.*, 2023), performance analysis in practical assessments such as OSCE and workplace-based assessments mediated by digital tools (Soledad *et al.*, 2025; Jeffery *et al.*,

2024), as well as the use of language models for clinical scenario simulation and assessment training (Sallam, 2023; Ayers *et al.*, 2023). Impacts on assessment reliability, standardization of faculty judgment, and personalization of learning were also discussed (Kassam *et al.*, 2024). Gaps were identified regarding pedagogical validation of systems, algorithm transparency, and measurement of longitudinal impact on real professional competence (Feigerlova *et al.*, 2025; Tozsín *et al.*, 2024).

Next, the synthesis of the studies included in this review is presented, organizing their main methodological characteristics and respective contributions to the investigated field.

Table 1: Core studies supporting the analytical categories of clinical competence assessment in medical education in the context of artificial intelligence

Article Title / First Author	Journal / Country	Language / Year	Method / Level of Evidence	Main Findings / Contribution
Ethical use of artificial intelligence in health professions education – Masters	Medical Teacher / UK	English / 2023	AMEE Guideline / Expert consensus	Establishes ethical framework for AI use in medical education and emphasizes transparency, authorship, and professional responsibility in AI-assisted tasks
ChatGPT utility in healthcare education, research, and practice – Sallam	Healthcare / Switzerland	English / 2023	Systematic Review / Level I	Identifies opportunities and risks of generative AI in learning and assessment, including threats to validity of written assignments
Performance of ChatGPT on USMLE – Kung	PLOS Digital Health / USA	English / 2023	Experimental performance analysis / Level III	Demonstrates AI capability to solve clinical reasoning questions, challenging traditional cognitive assessments
The role of artificial intelligence in medical education – Tozsín	Surgical Innovation / USA	English / 2024	Systematic Review / Level I	AI enhances personalized feedback and simulation-based learning but requires pedagogical validation
Utilization of AI among medical students – Shi	JMIR Human Factors / China	English / 2026	Cross-sectional study / Level IV	Shows widespread student use of AI tools and highlights need for assessment redesign
Expert consensus on AI proficiency for medical educators – Pan	Chinese Academy of Medical Sciences / China	English / 2026	Consensus guideline / Level V	Defines competencies required for educators to evaluate learners in AI-mediated environments
Using the OSCE to assess communication and reasoning – Soledad	BMC Medical Education / Spain	English / 2025	Longitudinal observational study / Level III	Confirms OSCE as valid performance-based assessment for clinical reasoning and communication
Entrustable professional activities and entrustment decision making – Ten Cate	Academic Medicine / USA	English / 2021	Conceptual framework / Level V	Provides theoretical basis for competence inference through workplace-based assessment

Recommended description of an EPA – Ten Cate & Taylor	Medical Teacher / UK	English / 2021	Guideline / Level V	Standardizes evaluation of professional performance and decision-making autonomy
Reimagining medical education in the age of AI – Wartman	Academic Medicine / USA	English / 2019	Perspective / Level V	Introduces paradigm shift from knowledge recall to cognitive partnership with AI

Source: Prepared by the authors based on the analyzed studies.

The table presents the studies that directly supported the analytical categories discussed in this review. As characteristic of integrative reviews, not all selected studies are individually described, but all contributed to the interpretative synthesis of evidence.

4 DISCUSSION

The results of this review demonstrate that the presence of artificial intelligence in medical education does not merely represent the incorporation of a new technological tool, but rather the emergence of a paradigmatic reorganization of assessment processes (Masters, 2023; Wartman & Combs, 2019). The assessment of clinical competence, historically based on direct observation of performance and inference of the student's clinical reasoning, now coexists with systems capable of simulating patients, suggesting diagnostic hypotheses, and structuring complete clinical responses (Kung *et al.*, 2023; Ayers *et al.*, 2023). In this context, the evaluative focus progressively shifts from verifying the correct answer to analyzing the reasoning construction process (Ten Cate *et al.*, 2021).

The literature does not indicate a replacement of the human assessor by technology, but rather the configuration of a new hybrid assessment ecosystem in which traditional instruments coexist with automated feedback and performance analysis resources (Sallam, 2023; Feigerlova *et al.*, 2025). This coexistence generates methodological and pedagogical tensions: while artificial intelligence expands possibilities for individualized monitoring, it also challenges the validity of assessment models based solely on the student's final output (Kassam *et al.*, 2024). Therefore, understanding the impact of artificial intelligence on assessment implies understanding changes in the very definition of clinical competence.

Based on these findings, the results were organized into two analytical axes: (1) reconfiguration of clinical competence assessment instruments and (2) ethical and epistemological challenges of AI-mediated medical assessment.

4.1 Reconfiguration of clinical competence assessment instruments

The studies demonstrate that classical instruments in medical education, such as OSCE, Mini-CEX, and workplace-based assessments, now operate in a context in which the student has immediate external cognitive support (Soledad *et al.*, 2025; Ten Cate & Taylor, 2021). Artificial intelligence modifies the nature of the assessed task: it is no longer merely the ability to recall and apply knowledge, but rather the ability to select, interpret, and critically appraise information produced by digital systems (Shi *et al.*, 2026). In this sense, clinical competence comes to involve metacognitive judgment rather than only technical performance.

In this new scenario, formative assessment gains centrality. AI-based tools enable immediate feedback, identification of individual learning gaps, and longitudinal monitoring of performance, expanding the pedagogical potential of assessment (Sallam, 2023). The literature indicates that such resources support the development of reflective clinical reasoning, provided they are integrated into structured faculty mediation (Kung *et al.*, 2023). Technology, therefore, does not replace the educational process but enhances its diagnostic capacity regarding learning.

However, risks also emerge. Some studies indicate the possibility of superficial performance when students rely on automatically generated responses without understanding them (Gilson *et al.*, 2023). In these cases, assessment instruments based solely on the final product lose validity. Thus, a necessary shift is observed from answer-centered assessment toward justification-centered evaluation, focusing on decision-making processes and the explicit articulation of clinical reasoning, reinforcing the interpretative nature of professional competence (Ten Cate *et al.*, 2021).

4.2 Ethical and epistemological challenges of AI-mediated medical assessment

The incorporation of artificial intelligence into medical training introduces questions regarding authorship, reliability, and authenticity of the assessed performance (Pan *et al.*, 2026). The possibility of automatically generated clinical management challenges traditional models of individual assessment, requiring a redefinition of what it means to demonstrate competence (Wartman & Combs, 2019). The literature indicates that the central issue is not the use of the tool itself, but the invisibility of its participation in the task-resolution process (Feigerlova *et al.*, 2025).

In this context, the role of the evaluator changes. The teacher ceases to be merely a judge of performance and becomes a mediator of clinical interpretation. Assessment comes to mean

investigating how the student uses artificial intelligence, what criteria are adopted to accept or reject its suggestions, and how such information is integrated into the real clinical context (Sallam, 2023). Clinical competence thus comes to include decisional responsibility and awareness of technological limitations.

Furthermore, ethical implications arise concerning patient safety and professional training. If future clinical practice will inevitably be mediated by intelligent systems, it becomes inadequate to assess students while ignoring their existence. However, evaluating while passively accepting their responses compromises professional autonomy (Izquierdo-Condoy *et al.*, 2026; Masters, 2023). The literature therefore points to the need for assessments that examine human-technology interaction rather than the student's isolated performance (Wartman & Combs, 2019).

5 CONCLUSION

This integrative review demonstrated that artificial intelligence does not merely represent the introduction of a new tool in medical education, but rather a shift in the very meaning of clinical competence. Assessment moves beyond verifying correct answers and comes to involve the analysis of professional judgment in relation to information generated by intelligent systems.

The studies indicate relevant formative potential, particularly through expanded feedback and individualized monitoring. However, the validity of assessment depends on maintaining the centrality of clinical reasoning and on structured faculty mediation, avoiding both prohibition and unrestricted acceptance of technological use.

Gaps remain regarding the pedagogical validation of new assessment formats and their longitudinal impact on professional practice. Therefore, the development of assessment models that explicitly integrate the critical use of artificial intelligence as a component of contemporary medical competence is recommended.

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